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# Motion-Based Consequence Analysis: From Pre-Operation Risk Assessment to Live Operational Monitoring

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## Abstract

Current DP rules and guidance specify capability analysis in detail but treat consequence analysis (CQA) as little more than a periodic alarm. The prevailing requirement — recompute every five minutes, raise a flag only once a limit is breached — leaves the operator with no continuous indication of the probability of exceeding operational limits, in the intact condition or following a worst-case failure (WCF), and no view of the motion consequences of such a failure before it happens. The risk that CQA is meant to manage is inherently dynamic; the rules treat it as static.

This work develops a motion-based CQA framework that closes that gap end-to-end. A pre-operation risk assessment extends the conventional capability analysis with an estimate of vessel response and post-WCF excursion from the planned sea state, allowing the worst-case failure design intent (WCFDI) to be evaluated in motion terms. The same model is reformulated for online execution, producing a continuously updated CQA panel that estimates intact and post-WCF station-keeping footprint directly from live measurements and the DP model, without a runtime sea-state input. The panel is further extended to motion-sensitive mission equipment such as walk-to-work gangways.

Validation against a nonlinear DP simulator is complete; validation against logged operations is in progress.

## Abbreviations and Definitions

**CQA** Consequence Analysis: the DP-rules-mandated periodic computation that flags whether the vessel can maintain position after a worst-case failure. Referred to throughout as the **classical CQA** where it needs to be distinguished from the extension proposed here.

**Motion-based CQA** The extension of the classical CQA proposed in this paper, which quantifies the **probability of exceeding operational limits**, in both intact and post-WCF conditions, continuously and in motion terms.

**WCF** Worst-Case Failure: the failure event (typically loss of a thruster group or bus section) against which CQA is evaluated.

**WCFDI** Worst-Case Failure Design Intent: the specific failure configuration the capability and consequence analyses are evaluated against; the vessel is designed to survive this.

**DP model** The mathematical model of the vessel used internally in the DP control system to estimate the vessel response from the combination of thruster action, sensor measurements, and position reference measurements. It carries estimates of position, velocity, and the slowly varying bias that absorbs the environmental load not otherwise modelled.

**Capability analysis** The classical pre-operation deliverable computing the achievable envelope of environmental force the vessel can hold against.

**WCFDI decision matrix** A traffic-light forecast of intact and post-WCFDI operability as a function of (vessel heading, forecast time), delivered before the operation starts.

## Introduction

Consequence analysis (CQA) has been a requirement for redundant DP classes for several decades. The IMO consolidated the requirement at international level in MSC/Circ.645 in 1994 [1], and it is carried into the class rules [2, 3] as a mandatory software function on DP-2 and DP-3 vessels. Its purpose is straightforward:

given the worst-case failure (WCF) the vessel is designed against, verify continuously that the DP system as a whole — sensors, thrusters, power, and control — can withstand the average prevailing environmental forces after the failure. If it cannot, an alarm is raised.

That obligation is met today by a periodic force-and-moment reserve check with alarm-on-breach. The check is defensible: applied with a percentage capacity margin it keeps the operator comfortably inside a safe envelope for the majority of operations. What it does not report is **motion**. The CQA does not indicate how the vessel will move in the intact condition, how it will move during and after a WCF, or how mission-sensitive equipment such as walk-to-work gangways, heave-compensated cranes, and ROV LARS will move with it. It does not quantify the probability of exceeding operational limits before those limits are breached. Modern DP operations increasingly run at sea states where the motion of the vessel or its mission equipment sits close to operational limits well inside the capacity-safe envelope, and the operator has no continuous, graded indication of proximity to those limits.

This paper proposes a **motion-based consequence quality analysis** (CQA) framework that closes that gap by adding a layer on top of the current CQA implementation, in two coupled deliverables: a **pre-operation risk assessment**, evaluating the worst-case failure design intent (WCFDI) in motion terms from the forecast sea state; and a **live CQA panel**, producing continuously updated $P_{50}$ / $P_{95}$ motion-footprint envelopes for intact and post-WCF conditions from measurements every DP-2/DP-3 vessel already publishes. The framework is additive: the current CQA continues to run and continues to raise its Drift Off / Force Off alarms; the motion layer adds one new operator alert, **Large Transient**, that fires when the motion prediction itself should be read as a lower bound rather than an upper bound. With the motion layer in place the operator can work closer to the operational limit in a controlled way, seeing proximity to the limit rather than waiting for a breach.

The remainder of the paper is organised as follows. Section 2 states the design goals of the framework. Section 3 develops the pre-operation motion capability analysis. Section 4 develops the live CQA panel and the two-channel design. Section 5 presents simulator-based validation on a CSOV benchmark vessel. Section 6 reports the in-progress validation against real operational logs. Section 7 concludes.

## Goal

The motion-based CQA framework is designed to two operational goals. Both are stated from the perspective of the operator planning and running an offshore operation — for example a walk-to-work personnel transfer or a heave-compensated construction lift — where the station-keeping capacity is comfortable but the **motion** of vessel and mission equipment may sit close to the operational limits.

1. **Verify motion limits before the operation.** Before committing to the operation, the operator can evaluate whether the planned vessel and mission-equipment motion — both in the intact condition and following the worst-case failure (WCF) — will stay inside the operational limits at the planned sea state, heading, and operating window. The classical capability analysis already verifies that the DP system has the capacity to hold; the motion-based pre-operation risk assessment answers the next question, whether the resulting motion is inside the limits that actually constrain the operation.
2. **Maintain continuous awareness of the safety margin towards those motion limits during the operation.** Once the operation is in progress, the operator continuously sees how close the vessel and mission-equipment motion sits to the operational limits, both in the intact condition and in the post-WCF condition. Rather than the binary alarm-on-breach of the classical CQA, the operator reads a continuously updated margin: comfortable, closer, at the limit. The awareness covers the post-WCF condition explicitly, so the operator sees whether the operation is inside the survivable motion

envelope, not only inside the capacity envelope.

Four design principles support these two goals and are worth stating briefly up front so the methodology sections that follow can be read against them. First, **motion-based throughout**: pre-operation and live outputs are stated in vessel and mission-equipment motion, in the same units and thresholds the operator already reads on the DP console. Second, **one model in two regimes**: the same underlying motion model serves the pre-operation stage (fed by the forecast sea state) and the live stage (fed by live measurements and the DP model), so that the operator sees the same numbers before and during the operation. Third, **readily available measurements only, online**: the live layer consumes only signals every DP-2/DP-3 vessel already publishes, plus vessel-constant parameters — no runtime seakeeping model, no runtime spectral partition, no runtime wave forecast — so that the design stays defensible under mixed wind-sea and swell, cross-current forcing, and near-installation shielding. Fourth, **additive to the classical framework**: the motion layer sits on top of the classical capability + CQA computation and consumes its outputs, without replacing or contesting any rule-mandated behaviour; the classical Drift Off and Force Off alarms remain the primary hard-failure signals.

## Motion-Based Capability Analysis

### Why motion matters in modern DP operations

Consider a walk-to-work operation transferring personnel to a wind turbine. The gangway operator has a firm limit on tip motion — a few tenths of a metre — beyond which the transfer is stopped. In many sea states the vessel itself is comfortable: the DP capacity margin against wind, wave, and current is generous, and the classical CQA reports green. Yet the transfer stops, because the **motion** projected through the gangway lever arms exceeds the tip limit long before the vessel exceeds its station-keeping limits. The same picture recurs in heave-compensated construction, ROV LARS operations, and offshore crane work: the operational limit is a motion limit on mission-sensitive equipment, and it sits inside the capacity-safe envelope reported by the classical CQA. Even for the vessel itself, the IMCA M254 [4] station-keeping alarm concept is inherently a motion concept: the yellow (P90) and red (P95) footprint bands are quantiles of the LF motion process, not a percentage of capacity reserve. A motion-based pre-operation risk assessment produces exactly these quantiles for the vessel and, through the same shared kinematic model, for any point on the mission equipment that the operator cares about.

### High-quality weather forecasts as an enabler

The motion-based pre-operation risk assessment is only defensible because the meteorological input is now good enough. Modern global ensemble products such as the ECMWF Integrated Forecasting System [5] deliver directional wave spectra with useful skill at 24 to 72 hours horizon for the wind-sea component, together with a probabilistic view from the ensemble spread. Regional hindcast reanalyses such as NORA3 [6] provide a directional spectrum climatology at 3 km resolution for the North Sea and Norwegian Sea that has been extensively validated against buoy measurements and is a useful cross-check against archived forecasts. Skill on the swell component and on rapidly evolving weather systems remains the harder part of the problem, and cross-basin swell propagation and mixed multi-modal sea states are known weak points; but the forecast directional spectrum is credible enough that using it as the input to a motion-response model produces an operationally meaningful envelope rather than a numerical artefact. The pre-op layer of the framework treats the forecast directional spectrum as its primary input and propagates it through the vessel response model described in the next subsection.

## Method: from forecast sea state to motion envelope

The pre-operation method takes the forecast sea state as input and produces a motion-footprint envelope — a probability distribution of peak excursion over the operating window — for both the vessel and any user-defined point on mission-sensitive equipment. It comprises five steps.

**a. Vessel-response model.** A 3-DOF linearised model of the DP station-keeping regime,

$$M\eta\ D\eta = \tau_{env}\ \tau_c, \tag{1}$$

with $\eta = x, y, \psi^\top$ the position and heading deviation from setpoint, $M$ the mass and added-mass matrix, $D$ the linear damping, $\tau_c$ the DP thrust command from the vessel's own PID

- integrator controller closed around $\eta$, and $\tau_{env}$ the

environmental forcing. Only vessel mass, damping, and controller tuning are required; a full 6-DOF seakeeping representation is not used because the pre-op deliverable is a horizontal-plane station-keeping footprint.

**b. Environmental forcing.** From the forecast directional wave spectrum, mean wind, and mean current, $\tau_{env}$ is composed of three standard components: the second-order wave-drift force per DOF, obtained by applying the vessel's quadratic transfer functions (QTFs) [7] to the directional spectrum; the wind force per Isherwood coefficients [8]; and the current force per OCIMF coefficients [9]. The wave-drift term dominates the LF driving spectrum; wind and current appear primarily as bias offsets absorbed by the DP model.

**c. From $\tau_{env}$ to motion envelope.** Under Eq. (1), the intact response has an LF component driven by the wave-drift and wind spectral peaks and a wave-frequency component driven by first-order wave excitation; both are approximately stationary Gaussian on the 30-min operating-window timescale. The station-keeping footprint reported by the framework is the running maximum $R$ of $|\eta_{xy}t|$ over that window, whose distribution is given by the Rice [10] / Vanmarcke [11] running-max formula in terms of the process variance $\sigma^2$, the zero-upcrossing rate, and a spectral bandwidth parameter. The operator display reports the median $P_{50}$ and tail $P_{95}$ of $R$; the traffic-light band that decides operable / caution / no-go is driven by $P_{90}$ against the vessel- and mission-specific warning and alarm radii, matching the IMCA M254 station-keeping alarm concept [4]. The wave-frequency component is added in quadrature.

**d. Extension to mission equipment.** The same excursion process, projected through mission-equipment lever arms and coupling kinematics, gives the motion at any point of interest — a gangway tip, LARS boom head, or crane hook — as a linear combination of vessel translations and rotations. No new random inputs are introduced; vessel and mission-equipment footprints appear together in the pre-op operator summary.

**e. Note on conservatism.** The Rice / Vanmarcke running-max formula assumes independence of successive up-crossings. Under feedback- controlled DP the controller compresses the LF tail, so the assumption is mildly conservative — typically about 1.5× on the P95 excursion. The framework treats this as safety margin; under-promising motion performance is preferred to over-promising it.

## Assumptions

The pre-operation motion model rests on the following explicit assumptions.

- The sea state is stationary over the operating window, so that process statistics (variance, spectral moments) are constant.

- The forecast directional spectrum is representative of the actual sea state during the window, within the skill of the forecast product used.
- The vessel QTFs, wind coefficients, and current coefficients are the design-intent values for the vessel and heading; drift and loading variation are absorbed into the framework's conservatism margin.
- Forward-speed effects are absent (station-keeping regime): the vessel is dynamically holding a setpoint, not transiting.
- Low-frequency and wave-frequency responses combine additively in quadrature into the running-maximum excursion.
- The classical capability analysis provides the outer envelope of operability. The motion-based envelope produced here is the inner envelope: within the capacity-safe outer envelope, the motion envelope identifies where operational limits — vessel or mission equipment — are first exceeded. The two envelopes are used together, not one instead of the other.

### Example: intact operability polar

Figure 1 illustrates the intact-condition operability polar produced by the framework for the CSOV benchmark vessel introduced in Section 5. At any given vessel heading and mean wind speed the polar answers whether the operation is operable (green), in a caution band (yellow), or no-go (red) in terms of the intact motion footprint. The colour band is driven by the P90 of the running-maximum excursion against a warning and alarm radius: for the vessel base position the warn / alarm thresholds are the IMCA M254 station-keeping levels (here 2.0 m / 4.0 m); for the gangway telescope they are the mission-equipment stroke limits (4.20 m / 5.60 m). The sea state at each polar radius is generated from the wind speed via a Pierson-Moskowitz relation with a fixed current component, and the operating window is 20 minutes.

Two features are worth calling out. First, the operable envelopes are visibly anisotropic: both panels show larger operable radii in head and following seas than in the beam sector, as expected from the projection of environmental force through the vessel geometry. Second, the gangway telescope panel has a larger operable envelope than the vessel base position, because the gangway threshold levels are set from the mechanical stroke of the telescope, not from a station-keeping alarm limit; the vessel and mission-equipment footprints are treated in the same motion-based framework but each against its own operational limits. The framework can produce the same polar for any additional mission-equipment point that the operator provides thresholds for.

### Example: post-WCFDI operability polar

Figure 2 overlays the post-WCFDI no-go boundary (black dashed) on the same intact-prior polar of the previous subsection. The WCFDI scenario for this illustrative figure is a loss of one of three symmetric thruster groups, which for the standalone prototype used to generate the figure is parameterised as a per-DOF surviving-fleet capacity of two-thirds of intact and an allocator reallocation time $T_{\text{realloc}} = 10$ s; the P90 Gaussian factor is $k_\sigma = 0.674$. In a real deployment the per-DOF residual capacity is not a scalar of intact — the motion layer consumes whichever $\tau_{\text{residual, cap}}$ the classical CQA already publishes over the survivor set. The dashed boundary is the wind-speed contour at which the P90 of the running-maximum excursion, evaluated on the post-WCFDI closed-loop dynamics with this residual capacity, reaches the alarm radius per DOF.

The important operational message is that the post-WCFDI boundary sits well inside the intact alarm ring — the vessel panel shows the WCFDI no-go contour at roughly 15-20 m/s wind speed in the beam sector,

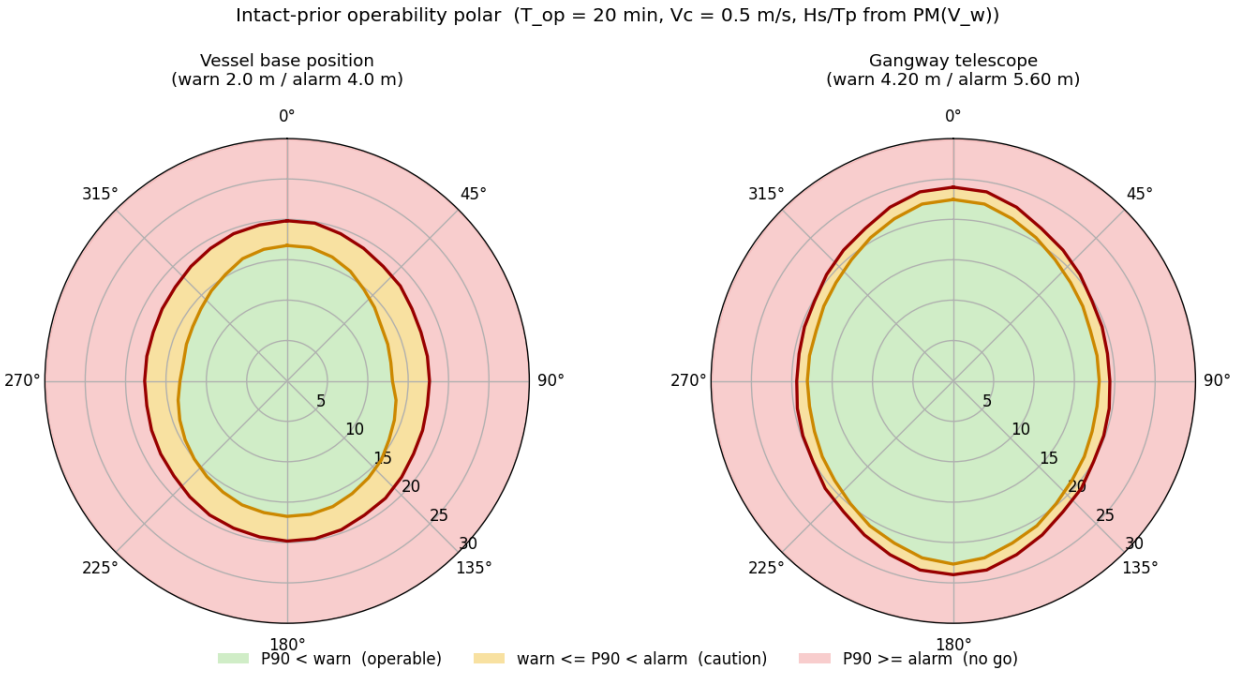



Figure 1: Intact-condition operability polar for the CSOV benchmark vessel and forward walk-to-work gangway. Polar radius is mean wind speed [m/s]; heading is vessel-relative wave direction. Colour band is driven by the P90 of the running-maximum motion excursion against the vessel / mission-equipment warning and alarm radii.

compared to the intact alarm at around 25 m/s. This is the motion equivalent of the tighter capacity envelope the classical CQA already screens against, but expressed in the operator's motion units rather than in a capacity-reserve percentage. Two features of the plot deserve comment. First, the WCFDI restriction is materially asymmetric across headings: the boundary is tightest in the beam sector (heading 90 / 270°) where the DOF that lost the most capacity is loaded worst, and closest to the intact envelope in head and following seas. Second, the WCFDI restriction bites less hard on the gangway telescope panel than on the vessel base position panel: the mission-equipment threshold is a mechanical stroke limit that is generous relative to the post-WCFDI excursion at the same sea state. For this vessel and gangway configuration the vessel-station-keeping metric is the binding constraint after the WCFDI, not the gangway telescope.

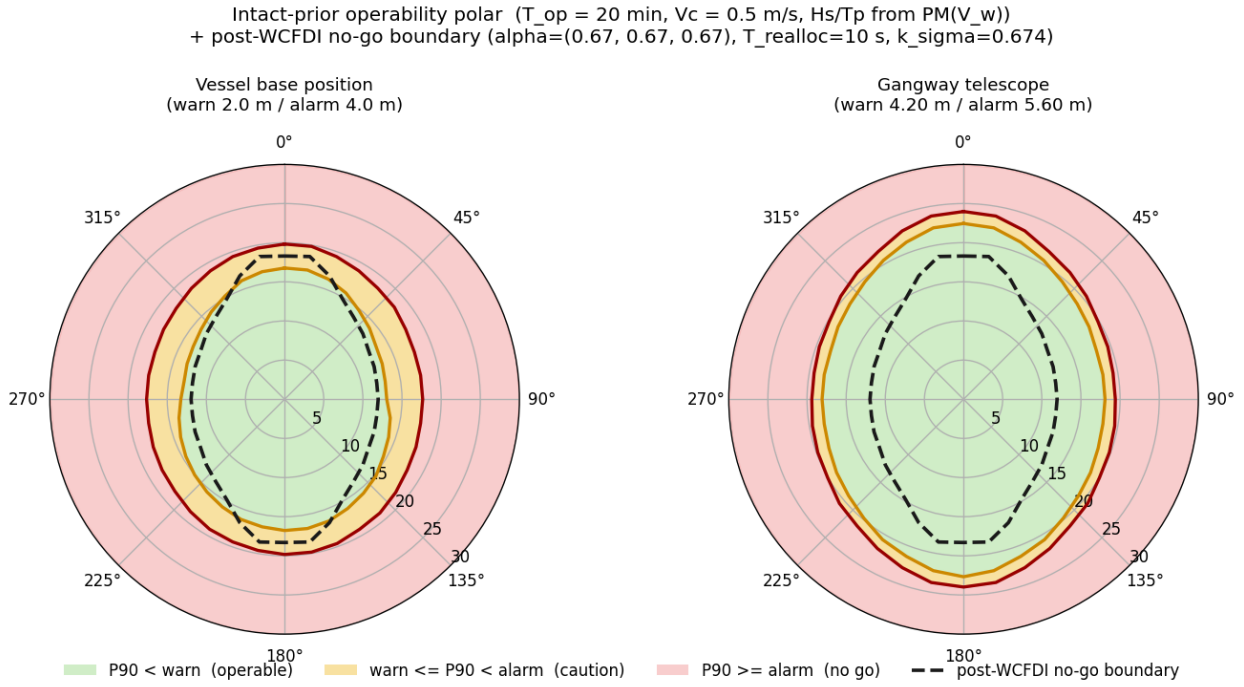



Figure 2: Post-WCFDI operability polar for the CSOV benchmark vessel and forward walk-to-work gangway. The coloured bands are the intact-condition operability of Figure 1; the black dashed contour is the wind-speed limit at which the post-WCFDI running-maximum P90 reaches the alarm radius. Scenario: loss of one of three thruster groups (per-DOF surviving-fleet capacity two-thirds of intact for this illustrative figure), allocator reallocation time 10 s.

## Example: WCFDI decision matrix

The operability polar of the previous subsection gives a static picture: for a given sea state, which headings are operable? The WCFDI decision matrix (Figure 3) puts that picture on a forecast timeline. Each cell of the matrix is one (heading, forecast slot) combination; the cell colour is the utilisation ratio evaluated on the forecast sea state at that time, using the same green→yellow→red gradient across all three panels so that different views of operability can be read directly against each other. The example is driven by a synthetic 24-hour storm.

The three panels correspond to three different consequence-analysis questions asked at the same (time, heading) cell:

- **Classical CQA utilisation** (top panel): the ratio of mean environmental force to residual (post-WCFDI) thrust capability, per DOF, taken maximum across DOFs. This is what the current force-and-moment reserve check that runs on any DP-2/DP-3 vessel reports. The operational bands used by the reference DP system are caution at 80% utilisation and warning at 90%.
- **Motion-based intact utilisation** (middle panel): the ratio of the intact-condition P90 running-maximum footprint to the IMCA M254 alarm radius. Reaches 1.0 when the intact motion of the vessel itself would cross the M254 alarm threshold.
- **Motion-based post-WCFDI utilisation** (bottom panel): the same ratio computed on the post-WCFDI motion envelope, i.e. what the motion would be if the worst-case failure occurred at that (time, heading) cell.

The operator's task is to plan a safe operation through the forecast window. The three panels answer that task in three different ways. The classical CQA panel identifies the operational constraint the current rules already enforce: a narrow yellow-orange caution band and a small red spot at the storm peak (12:00) in the two beam sectors, driven by the environmental force approaching the residual capability. An operator running only the classical CQA would read this as "the storm peak is borderline in beam headings; work around it and the operation is fine". The motion-based intact panel adds essentially nothing new here — the vessel motion itself never comes close to the alarm radius, so the classical CQA plus intact motion confirms the same operational verdict.

The motion-based post-WCFDI panel changes the picture. Even though the classical CQA reports at most yellow-caution across the storm peak, the motion-based post-WCFDI utilisation reaches saturation (dark red, well past the alarm radius) in a narrow ~2-hour window around 12:00 in the beam headings. Physically, this is the regime where the storm energy is still comfortably below the vessel's survivable capacity but the **motion** consequence of a WCF at that instant would take the vessel outside the operational motion envelope. Two heading windows around head and following seas (roughly 150-210° and 330-030°) stay green even at the storm peak.

The operator's plan therefore depends on which panel is authoritative. On classical CQA alone the plan is "operate through the storm, easing off around 12:00"; on the motion-based post-WCFDI panel the plan becomes "avoid the beam sector 11:00-13:00 window entirely, or use one of the two green heading windows through it". The 2-hour narrow red band is the added operational content the motion-based framework delivers: the **when** and the **which heading** at the resolution the motion actually constrains, not just the **whether** at the resolution the residual capacity constrains.

### Example: pre-op operator summary

The polar plots and the decision matrix are planning views over sea state and time. The pre-operation operator summary (Figure 4) collapses a single operating point — the state selected for the operation — into a single-page display that the operator reads before committing to the operation. Two bars stack vertically: the top bar is vessel base-position capability, the bottom bar is gangway telescope capability, both against their own IMCA M254 warn / alarm bands. Each bar shows the median (open circle, $P_{50}$) and the tail (filled diamond, $P_{95}$) of the post-WCF running-maximum motion excursion evaluated on the current operating state, with the traffic-light background driven by the same P90-based threshold used in the polars.

The important content on the display is not the traffic light itself — the operator already reads that on the polar — but the two headline probability numbers on each bar: "P(footprint > amber)" and "P(footprint > red)". In the example shown, the vessel bar reports "if WCF now: P(footprint > 2 m) = 5.0 %, P(footprint > 4 m) = 0.0 %". This is the concrete delivery of the abstract's **probability of exceeding operational**

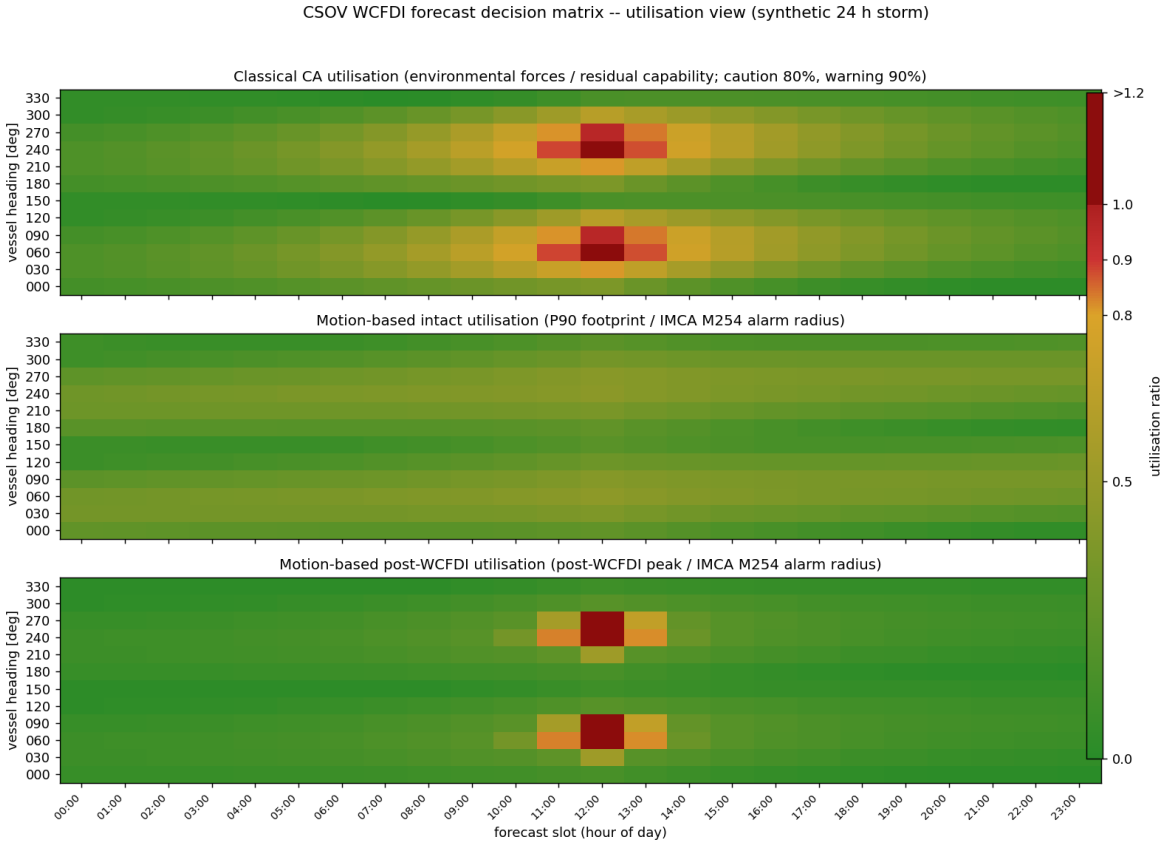



Figure 3: WCFDI forecast decision matrix over a synthetic 24-hour storm for the CSOV benchmark vessel, shown as three side-by-side utilisation heatmaps. Rows: vessel heading (deg, true). Columns: forecast slot (hour of day). All three panels use the same green→yellow→red gradient colour map on a common utilisation ratio scale 0 to 1.2 (colour bar right). Reference thresholds: for the classical CQA panel, caution at 0.80 and warning at 0.90 (reference DP system convention); for the two motion panels, caution at 0.50 (IMCA M254 warn / alarm ratio) and warning at 1.00 (P90 equals the M254 alarm radius). Top: classical CQA utilisation (environmental forces / residual capability, max across DOFs). Middle: motion-based intact utilisation (P90 footprint / IMCA M254 alarm radius). Bottom: motion-based post-WCFDI utilisation (post-WCFDI peak / IMCA M254 alarm radius).

**limits**: not "the vessel can hold" (yes), not "the footprint is 2 m" (a single point estimate), but a probabilistic statement that the operator can weigh against the operational acceptance criterion for the specific mission. The gangway bar reports the analogous "P(end-stop hit)". Both are direct products of the same motion-based framework used to build the polars and the decision matrix; the operator sees one coherent set of numbers before and during the operation.

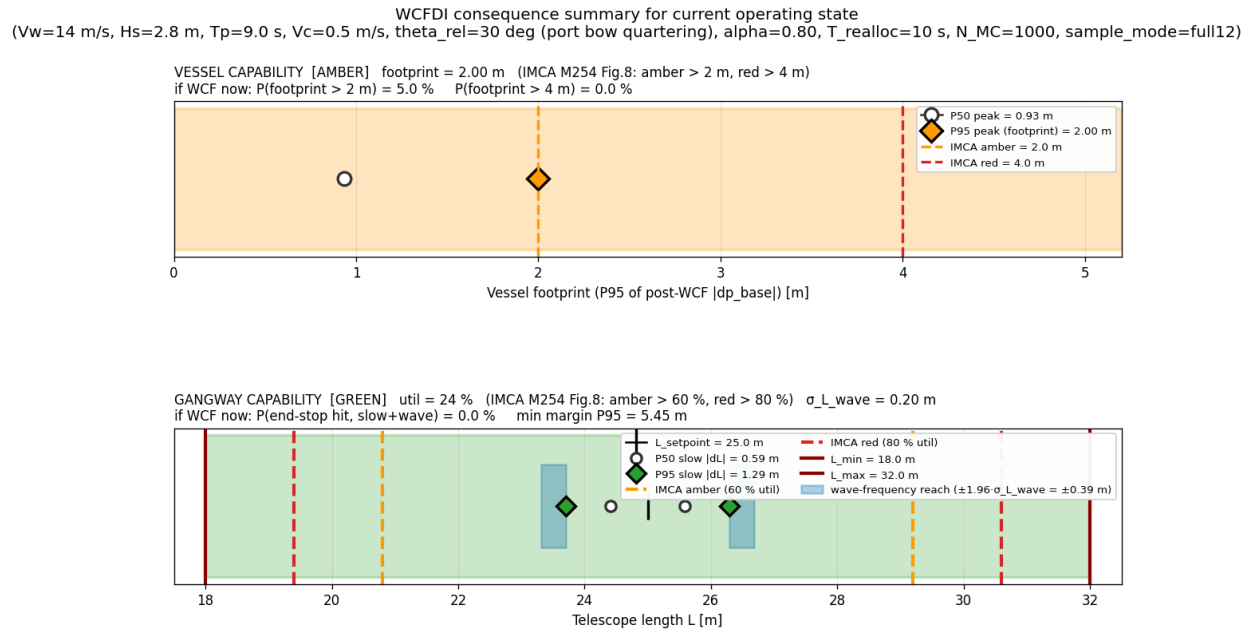



Figure 4: Pre-operation operator summary for the CSOV benchmark at a single operating point (Vw=14 m/s, Hs=2.8 m, Tp=9.0 s, bow-quartering seas, WCFDI scenario per Fig. 2). Top bar: vessel base-position capability. Bottom bar: gangway telescope capability. Open circle: $P_{50}$. Filled diamond: $P_{95}$. Vertical dashed lines: IMCA M254 warn / alarm thresholds per system. Headline probabilities quantify the risk of exceeding each threshold under the post-WCF motion excursion.

## Consequence Analysis (Live)

### Positioning: additive to the classical CQA

The live CQA panel runs alongside — not in place of — the classical CQA required by IMO MSC/Circ.645 [1]. The classical CQA continues to compute and publish four quantities that the motion layer subscribes to: the per-WCF-candidate capacity reserve, a drift-off flag per DOF (the survivor set has zero capability against the prevailing environmental force in that DOF), a force-off flag per DOF (the survivor set is available but the prevailing demand exceeds its cap), and $\tau_{\text{lost}}$, the vector of thrust that would be lost in the WCF at each candidate failure. The motion layer treats these as its inputs, along with the DP model state (position and velocity deviation $\eta, \eta$; bias estimate $\hat{b}$) and a rolling buffer of the allocator feedback $\tau_{\text{fb}}$; it does not duplicate the capacity-reserve computation and it does not consume any sea-state input at runtime.

$\tau_{\text{lost}}$ is not an instantaneous loss that stays at zero: the survivor set is reallocated back toward the pre-failure demand as fast as the thruster inertia and the allocator update rate allow. Figure 5 sketches the profile the motion layer assumes. At the WCF event, the available thrust capability magnitude drops instantaneously by

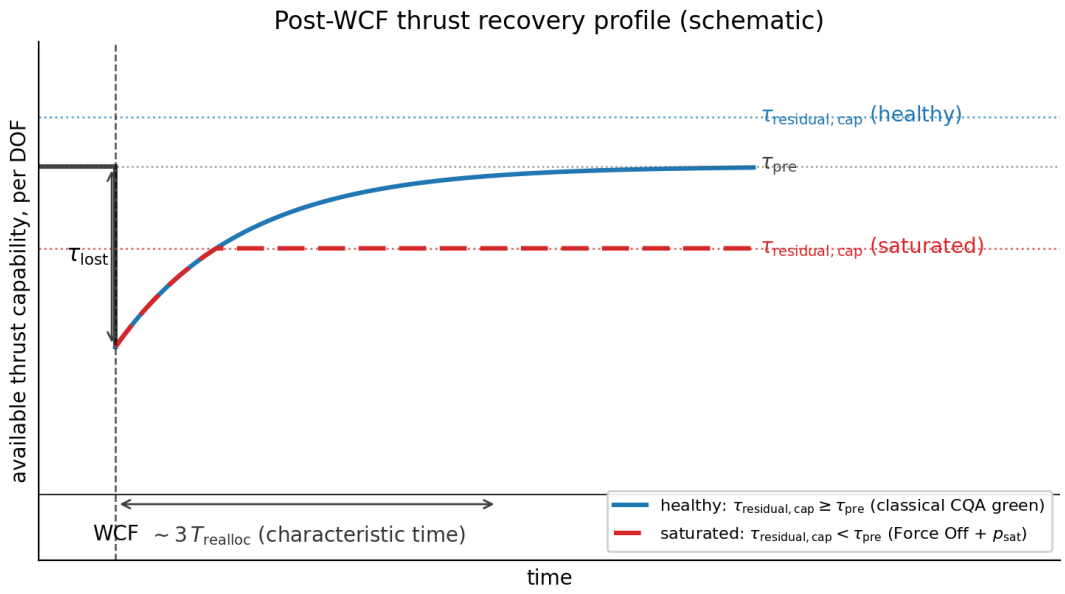



Figure 5: Post-WCF thrust recovery profile assumed by the motion layer, per DOF. At the WCF event the total commanded thrust capability drops instantaneously; the survivor set then recovers exponentially with characteristic time $T_{realloc}$, asymptoting either to the pre-failure demand $\tau_{pre}$ (healthy case, blue) or to the residual capacity cap \(\tau_{{residual, cap}}\) when $\tau_{pre}$ exceeds the cap (saturated case, red-dashed). The saturated case is the operational territory of the classical Force Off alarm and of the motion layer's saturation-probability channel.

$\tau_{lost}$, of which an immediately recovered fraction $\gamma_{imm}$ is picked up straight away by the survivor set. From that step the survivor allocator then recovers the remaining shortfall exponentially with characteristic time $T_{realloc}$, asymptoting to one of two targets. In the **healthy** case, when the pre-failure demand $\tau_{pre}$ is inside the residual capacity cap $\tau_{residual,\,cap}$ — the per-DOF thrust envelope of the surviving fleet, published by the classical CQA — the curve recovers back to $\tau_{pre}$ and the classical CQA reports green. In the **saturated** case, when $\tau_{pre}$ exceeds $\tau_{residual,\,cap}$, the same exponential recovery saturates at the cap instead of at $\tau_{pre}$; the classical CQA raises Force Off on that DOF, and the motion layer's saturation-probability channel raises the **Large Transient** alert on the transient itself. The motion layer consumes $\tau_{lost}$ and $\tau_{residual,\,cap}$ as-is from whatever classical CQA is running in the vessel — no vessel-specific capacity scalar is introduced. The one modelling parameter internal to the motion layer is the reallocation characteristic time $T_{realloc}$; for the CSOV benchmark used throughout this paper we take $T_{realloc}$ = 10 s, consistent with modern DP allocators. This time profile is what drives both the ensemble evaluator and the transient channel.

An important reading of Figure 5 is that it plots the **available** thrust capability per DOF, not the thrust the DP controller commands. The DP controller can and does command transient forces above $\tau_{pre}$ to arrest the motion picked up during the drop, provided the survivor set has the capability to deliver them; that commanded-thrust overshoot is captured inside the closed-loop vessel model of the method section (the controller term $\tau_{ctrl}t$ in Eq. 1), not in the recovery curve here. The first-order form of the recovery envelope is a compact approximation of the survivor-allocator dynamics: the immediately recovered fraction $\gamma_{imm}$ and the characteristic time $T_{realloc}$ are calibrated against reference DP simulator runs, and the approximation is confirmed to be operationally sufficient by the simulation validation.

The rule-mandated Drift Off and Force Off alarms remain the primary hard-failure signals in the operator UX; nothing in the motion layer suppresses or replaces them. The motion layer adds one new operator-visible

alert, **Large Transient**, that fires when the live allocator feedback approaches the survivor's cap and the linear motion prediction should be read as a lower bound rather than as a tight upper bound. The alert is designed to sit between the classical green state and the classical Force Off warning: it draws attention to the **approach** to a limit rather than to its breach. The mechanism is developed in the saturation-probability channel.

### Extreme-value method for exceedance envelopes

The intact-side extreme-value machinery on the live panel reuses the Rice / Vanmarcke running-maximum formula introduced in the method section [10], [11]; what is new here is the online source of the process moments $\mu, \sigma$ per DOF that the formula consumes.

The panel runs an online Bayesian estimator of the position-deviation standard deviation $\sigma$ on a sliding 10-minute window of the DP model's position-deviation channel $\eta t$. The estimator is recursive and produces a posterior distribution over $\sigma$ rather than a point estimate; the running-max evaluation is done at the posterior mean and cross-checked against the posterior 95th percentile so that early-warm-up transients on the estimator are visible to the operator as a widened envelope. Because the DP position-deviation samples are autocorrelated at the sample rate, the naive sample count over the sliding window would overstate the information available and produce an over-tight posterior. The estimator applies a Bartlett-style effective-sample-count correction that decouples the posterior variance from the sample rate and pins it to the process autocorrelation time.

The moments produced by the estimator feed the same running-maximum formula used offline: the panel reports $P_{50}$ and $P_{95}$ of the running maximum of $|\eta_{xy} t|$ over the next 30 minutes, against the same vessel and mission-equipment warn / alarm radii used in the pre-op operator summary. The sub-Rayleigh conservatism noted in the method section applies unchanged: the running-max formula assumes independence of up-crossings and is mildly conservative under feedback control, which is retained here as safety margin. The intact-side output of the live panel is therefore directly comparable to the pre-op operator summary, but continuously updated on live data and with no sea-state input at runtime.

### Motion after WCF: ensemble evaluator

The live panel needs a post-WCF motion prediction that is continuously updated on the vessel's actual state and can be trusted under the range of pre-failure conditions the WCF could catch it in. A single deterministic run of the vessel model from a point estimate of its steady state would collapse that range to a single number; the ensemble evaluator instead samples it. The idea is illustrated in Figure 6: rather than one trajectory, the panel propagates many, each starting from a recent snapshot of the actual vessel state, and reads its output from the spread across that ensemble.

Concretely, the evaluator maintains a rolling 300-second buffer of vessel-state snapshots $\left(\eta, \eta, \hat{b}, \tau_{\text{pre}}\right)$ drawn from the DP model and the allocator feedback. At each panel update, every snapshot in the buffer is used as the initial condition of the same closed-loop 3-DOF vessel model developed for the pre-op prediction in the method section (Eq. 1), propagated forward through the post-WCF window with $\tau_{\text{lost}}$ taken from the classical CQA output. Each propagation produces a peak-$|\eta_{xy}|$ trajectory; across the buffer this gives an empirical distribution of post-WCF peak excursions covering the actual pre-failure state variability the vessel is experiencing right now. The panel reports the median $P_{50}$ and the tail $P_{95}$ of that distribution, directly comparable to the pre-op operator summary.

Using the same vessel model the pre-op layer uses is deliberate: it enforces the "one model, two operating regimes" design goal of the Goal section. The two sides differ only in what feeds them — the pre-op side

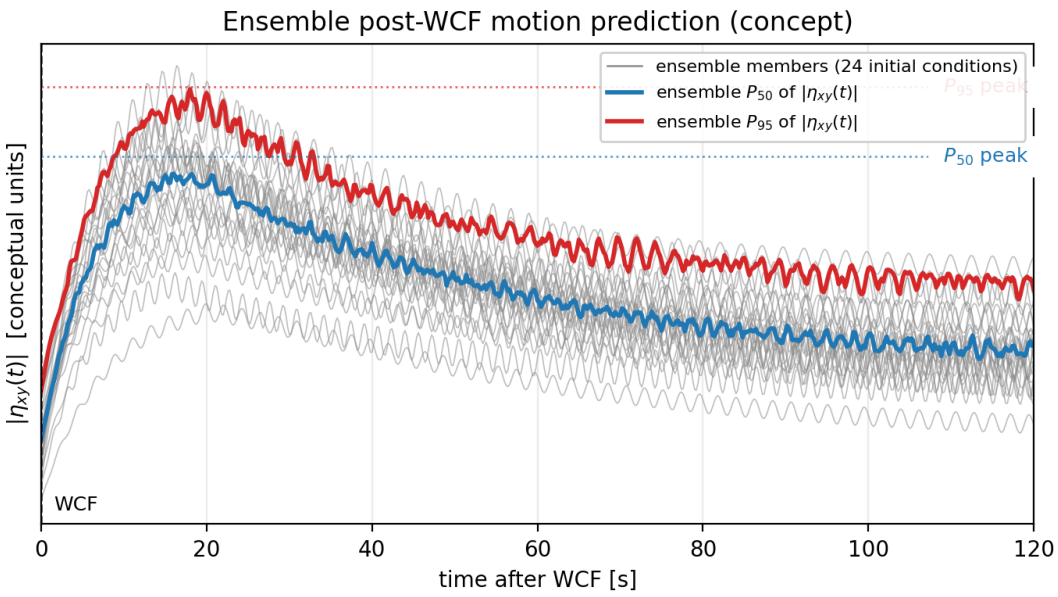



Figure 6: Conceptual illustration of the ensemble post-WCF motion prediction. At each panel update, snapshots of the recent vessel state are used as initial conditions for the vessel model, and each is propagated forward through the post-WCF window. The spread across the resulting trajectories is summarised by the ensemble $P_{50}$ and $P_{95}$ of $|\eta_{xy}t|$; the panel reports the $P_{50}$ and $P_{95}$ of the per-trajectory peaks (dotted lines). Numeric scales are omitted deliberately — the figure is illustrative of the mechanism, not of a specific vessel or sea state.

takes a forecast sea state through a vessel-response model to build the $\tau_{\text{env}}$ history; the live side takes the live $\hat{b}$ and $\tau_{\text{pre}}$ samples directly, without ever inferring a sea state. The classical CQA outputs $\tau_{\text{lost}}$ and the residual capacity feed both sides in the same way.

The ensemble evaluator handles the **long-term** post-WCF regime well because the closed-loop vessel model is quantitatively accurate for the non-saturated portion of the response. When the post-WCF residual allocator saturates, the linear model is no longer sufficient, and the motion prediction becomes a lower bound. That case is handled by the saturation-probability channel developed next.

### Transient right after WCF: saturation probability channel

The linear closed-loop vessel model that drives the ensemble evaluator is quantitatively accurate as long as the post-WCF residual allocator does not saturate. Once the survivor set is forced onto its capacity cap, the linear model breaks down and the ensemble output becomes a lower bound on the true motion — no longer a tight prediction, and therefore no longer safe to display as the operator's motion number. The saturation-probability channel is an independent live gate that tells the operator when this happens.

The mechanism is direct. The panel buffers the live allocator feedback $\tau_{\text{fb}}t$ on a rolling 300-second window and low-pass-filters out the wave-frequency component to isolate the LF part that the residual allocator would have to absorb after a WCF. For each DOF a mean $\mu$ and standard deviation $\sigma$ are computed, and a saturation probability

$$p_{\text{sat}} = \max_{\text{DOF}} \Pr\left(|\mathcal{N}\mu, \sigma| > \tau_{\text{residual, cap}}\right) \tag{2}$$

is evaluated against the per-DOF residual cap $\tau_{\text{residual, cap}}$ computed from the classical CQA output. The panel raises the Large Transient alert at operator-configured amber and red thresholds on $p_{\text{sat}}$, matching the IMCA M254 Fig.8 caution / warning band pattern used elsewhere in the operator UX.

The semantics of the alert are precise. $p_{\text{sat}}$ does not answer **what is the vessel's excursion after a WCF**; it answers **how likely is it that the linear motion prediction can be trusted**. When $p_{\text{sat}}$ is in the green range, the output of the ensemble evaluator is a tight upper bound on the post-WCF excursion and is what the operator should act on. When $p_{\text{sat}}$ crosses into amber or red, the ensemble output should be read as a lower bound and the operator should treat the actual excursion as materially larger than the number on the screen. The channel therefore behaves as a **trust indicator on the motion prediction**, not as a duplicate of the classical CQA capacity check.

The channel closes the "readily available measurements only" design goal of the Goal section: it consumes only the live allocator feedback and the per-DOF residual cap constant computed from the classical CQA output. It does not consume a sea state, a wave forecast, or a spectral partition. This makes it defensible under the operationally common conditions — mixed wind-sea and swell, cross-current forcing, shielding — that would break a QTF-based online estimator of the post-WCF forcing.

### Two-channel operator decision surface

The ensemble evaluator and the saturation-probability channel are combined with the classical CQA output into the operator decision surface of Table 1. The table is the operator's mental model of the live panel in one view: for any combination of the three channels the operator has an unambiguous reading of what to trust and what to act on. The **Post-WCF motion** column reads the output of the ensemble evaluator relative to the intact P95: **similar** means the post-WCF prediction is close to the present intact footprint; **elevated** means it is materially larger. The **Saturation** column is the saturation-probability channel, which qualifies how far the motion number in the previous column can be trusted.

Table 1: Live operator decision surface. Rows correspond to increasing operational concern.

| Classical CQA | Post-WCF motion | Saturation | Interpretation |
|---|---|---|---|
| OK | similar | green | Vessel operating in a comfortable state; no action needed |
| OK | elevated | green | Elevated post-WCF motion, but the prediction is trustworthy |
| OK | elevated | amber | Motion prediction is a lower bound; treat with caution |
| OK | elevated | red | Motion prediction unreliable; treat as worst-case |
| Drift Off/Force Off | (any) | (any) | Classical alarm dominates; motion-layer output is context |

The decision surface preserves the classical alarm ordering: when the classical CQA fires Drift Off or Force Off, the classical alarm takes precedence and the motion-layer output is displayed as supporting context rather than as the operator's primary action cue. Within the classical-CQA-OK state, the two-channel readout gives the operator four distinct positions on the way toward saturation. The three operator-visible alerts of the framework are therefore:

- **Drift Off** (classical CQA warning) — unchanged from current DP-2/DP-3 practice.
- **Force Off** (classical CQA warning) — unchanged from current DP-2/DP-3 practice.
- **Large Transient** (motion CQA caution / warning) — new; fires when $p_{\text{sat}}$ crosses its amber or red threshold, indicating that the ensemble output should be read as a lower bound.

No other alerts are added or removed; existing DP UX conventions, including audible / visual behaviour of Drift Off and Force Off, are preserved.

### Long-term post-WCF motion larger than the short-term transient

The post-WCF response has two distinct phases with independent risk profiles. The **transient phase** — the first 30-60 s after the failure — is dominated by the survivor allocator absorbing the step in $\tau_{\text{lost}}$; its peak excursion is what the classical CQA implicitly verifies against by requiring the survivor set to generate the pre-failure force and moment. The **long-term phase** is a new steady state under reduced capacity: with the surviving peak $\tau$ closer to the mean environmental forcing than in the intact condition, the LF headroom against wave-drift variability is compressed and the running-maximum footprint grows.

In some sea states the long-term P95 footprint exceeds the transient peak. The classical CQA sees the transient pass and reports green, but the operator is nonetheless outside the motion envelope. The ensemble evaluator captures this correctly: it propagates the closed-loop vessel model over a window long enough (~5 min) to cover both phases, so the reported P95 is the true post-WCF peak, transient or long-term whichever is larger. This is the regime where the motion layer is materially different from a transient-focused CQA extension, and it is one of the cases exercised in the long-term post-WCF validation.

### Estimator convergence

The intact-side envelope and the transient channel both consume sliding-window estimates of process moments from live data, so the numbers on the operator screen are only as trustworthy as those estimates. Figure 7 shows the estimator settling behaviour on a stationary synthetic position-deviation channel with known true $\sigma = 0.60$ m. Three independent realisations are run through the same online estimator used on the live panel, with a prior deliberately biased high at $\sigma = 0.90$ m to make the warm-up visible. The 5-95% posterior credible band collapses onto the true value as data accumulates in the 10-minute window, and by the time the window is full the run-to- run spread is well inside the band width the operator sees on screen. The band width is the operator's cue: while it is wide the live panel numbers should be read with the extra margin the band implies; once it has tightened the numbers can be acted on as if they were the offline pre-op prediction.

The 10-minute window and the accompanying implicit assumption that the process is stationary over that window are the price of running the estimator online without a sea-state input. Both assumptions are comfortable against how sea states actually evolve in the North Sea and comparable basins: significant wave height, peak period, and mean wind speed on a typical forecast grid change on timescales of hours, not minutes, so a rolling window an order of magnitude shorter than the forcing decorrelation time captures a locally stationary process to good approximation. Frontal passages and squall lines are the practical exceptions and are already handled outside the CQA loop by conventional weather routing.

## Simulation Studies

### Test vessel and simulator

The framework is validated against a benchmark CSOV ($L_{pp} = 101.1$ m, $B = 22.4$ m, $T = 6.5$ m) configured with a five-thruster layout typical of the class: two bow tunnel thrusters, a bow azimuth, and two stern main-propeller / rudder units. The reference truth is a high-fidelity nonlinear DP simulator with seakeeping-derived hydrodynamics, per-thruster force and moment allocation including allocator dynamics, and a nonlinear passive observer of the same class used in fielded DP-2/DP-3 systems. The simulator produces vessel motion at the level required to verify a live CQA panel: the position deviation $\eta t$, velocity $\eta t$, allocator feedback $\tau_{\text{fb}} t$, and observer states $\hat{b} t$, $\hat{\eta}_w t$ that the panel would read on a real vessel are all available for direct comparison against the panel's own predictions. The panel is exercised on the same input streams the field

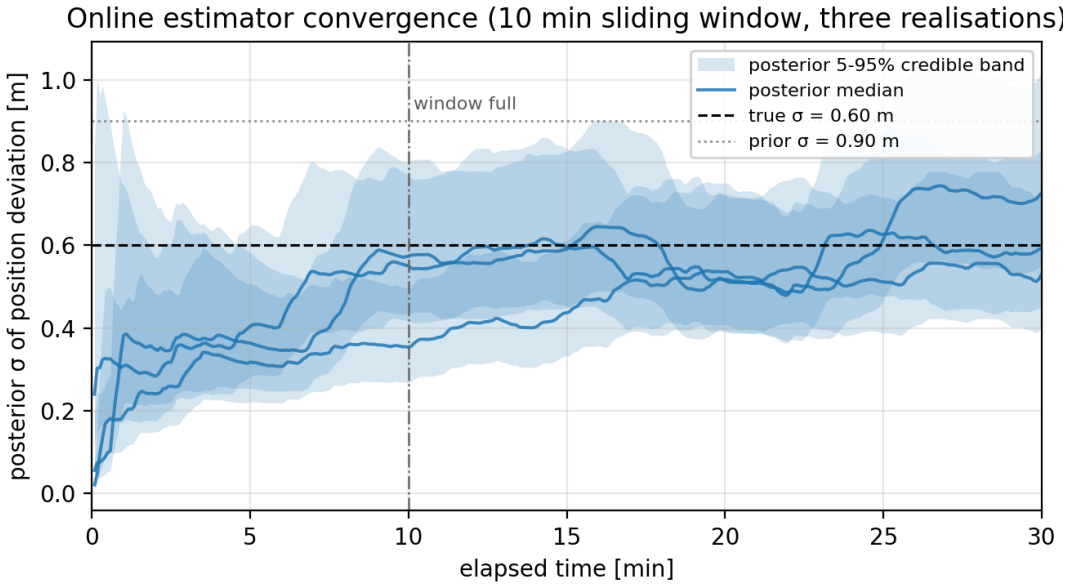



Figure 7: Online estimator convergence on a stationary synthetic position-deviation channel. Three independent realisations with true $\sigma = 0.60$ m are fed into the sliding 10-minute Bayesian estimator with a prior deliberately biased high at $\sigma = 0.90$ m to make warm-up visible. The posterior median (solid) and 5-95% credible band (shaded) converge onto the true value as the window fills; the band width itself is what the operator reads as trust in the live number.

system would see, with the simulator supplying the reference truth.

### Capability to intact motion

The left-hand panel of the motion roll-up (Figure 8) compares the framework's predicted $P_{95}$ of the intact-condition running-maximum footprint $|R_{LF}|$ against the corresponding simulator truth on each cell. The cells span Beaufort 4 to Beaufort 8.5 across head, quartering, and beam headings, plus two waves-only variants. Across all 14 cells the framework's $P_{95}$ is within +7% to +28% of the simulator truth, with a median error of +17%. All 14 cells are on the positive side of zero — the framework systematically overpredicts.

The systematic conservatism is not a modelling error; it is the sub-Rayleigh margin discussed in the method and extreme-value sections. The running-max formula assumes statistical independence of successive up-crossings, which is a mild overstatement under feedback-controlled DP because the controller compresses the LF tail. The magnitude of the observed bias (roughly a factor 1.15-1.20 in $P_{95}$) is consistent with the theoretical bound, and the sign is what the framework's design choice intended: the operator receives a mild upper bound rather than a mean or a lower bound.

The 14-cell agreement therefore validates the vessel-response model (3-DOF closed-loop dynamics with the observed DP integrator) and the extreme-value envelope (Rice / Vanmarcke running-max applied to the LF + WF process moments) on the intact side. The framework's intact-side output is the same object used in the pre-op operator summary and in the live intact bar of the extreme-value section; the roll-up confirms both.

### Transient at WCF

The right-hand panel of Figure 8 shows the framework's predicted post-WCF $P_{95}$ alongside the simulator truth on the same 14 cells. On the eleven cells where the survivor allocator does not saturate (the $p_{sat}$ channel remains green throughout), the framework's $P_{95}$ is within -23% to +16% of the simulator truth, with a median error of -10%. On these cells the linear closed-loop ODE that drives the ensemble evaluator is quantitatively accurate: the transient step in $\tau_{lost}$, the allocator absorbing that step, and the LF re-equilibration are all captured within the intrinsic scatter of the sea-state statistics.

The mild negative bias on the transient axis has a different origin than the positive bias on the intact axis. Where the intact axis sits on the sub-Rayleigh conservatism of the running-max formula, the WCF axis reflects the sea-state moment averaging over the 5-minute post-WCF window used by the ensemble evaluator. The integrated bias is small enough — comparable to the seed-to-seed scatter within each cell — that the framework's WCF number is a tight prediction rather than an upper or lower bound. Where the prediction should be treated as a lower bound is the regime the saturation-probability channel is designed to flag, and which the roll-up covers in the long-term post-WCF validation.

### Long-term post-WCF envelope

The rightmost columns of Figures 8 and 9 are the cells where the survivor allocator saturates in the post-WCF steady state — the regime the case-study of the long-term post-WCF discussion identifies as the operationally interesting one. On these cells the linear closed-loop ODE that drives the ensemble evaluator is no longer sufficient, and the framework's motion $P_{95}$ drifts from the simulator truth. The worst case in the roll-up is $\mathrm{bf8p5}_{\mathrm{q10w45}}$: the simulator reports an LF peak $P_{95}$ of 29.9 m, driven by seeds where the vessel loses station outright in the post-WCF long-term regime; the framework's predicted $P_{95}$ is 5.4 m, an 82% underprediction.

The 5.4 m motion prediction is nonetheless not what the operator sees on that cell. The framework raises a red alarm on $\mathrm{bf8p5}_{\mathrm{q10w45}}$ through two independent channels. First, the predicted motion $P_{95}$ of 5.4 m already exceeds the alarm radius of 4 m used for the CSOV, so the classical CQA motion bar fires red on 28 of 30 simulator seeds. Second, the saturation-probability channel evaluates $p_{sat}$ with a mean of 0.14 and a maximum of 0.26 on the same cell — well above the red threshold of 0.10 — and fires the Large Transient alert on 21 of 30 seeds and the amber threshold on the remaining 9. The operator therefore sees red on this cell through two independent gates before the simulator ever runs the WCF that would produce the catastrophic excursion.

This is the design intent of the two-channel layout of the operator decision surface: where the linear ODE breaks, the trust indicator triggers so that the operator is not misled by an artificially small $P_{95}$. The rejected-extensions section discusses three plausible refinements to the linear ODE that would in principle produce a tighter $P_{95}$ on the saturated cells, and were investigated and rejected.

### Sea-state and heading sweep summary

Figures 8 and 9 give the whole 14-cell scoreboard for the live CQA panel against the simulator truth. The cells cover Beaufort 4 through Beaufort 8.5 in head, quartering, and beam headings, plus two waves-only variants representative of purely-wave-driven station-keeping. Figure 8 covers the two motion axes the operator reads directly; Figure 9 covers the saturation channel that qualifies them.

The two operational messages of the roll-up are already stated in the intact-motion, transient, and long-term post-WCF validation sections, and are summarised here. First, on the 11 non- saturated cells the intact and post-WCF motion $P_{95}$ are accurate to within a factor of about 1.2 in either direction, with systematic biases

(positive on the intact axis, negative on the WCF axis) that reflect the intended design conservatism and the finite- window averaging respectively. Second, on the three cells where the survivor allocator approaches or crosses saturation (bf7p5$_{q10w45}$, bf8$_{q10w45}$, bf8p5$_{q10w45}$), the WCF motion $P_{95}$ underpredicts the simulator truth by up to 82% — but the two independent gates of the operator decision surface both fire red, giving the operator the correct actionable outcome despite the imprecise motion number.

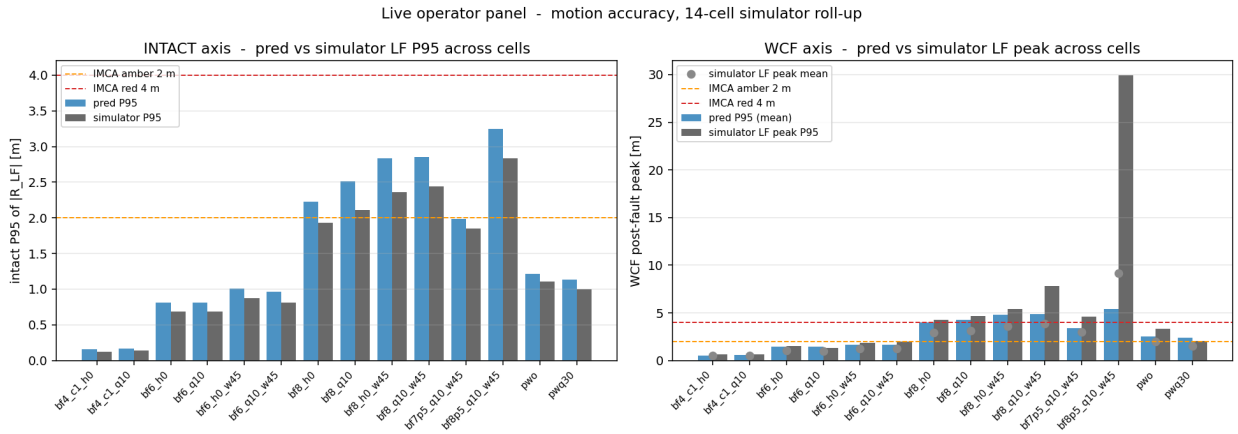



Figure 8: Motion accuracy across the 14-cell simulator roll-up. Left: intact axis $P_{95}$ of $|R_{LF}|$, predicted (blue) vs simulator (grey), against the IMCA amber (2 m) and red (4 m) alarm radii. Right: post-WCF $P_{95}$, predicted (blue) vs simulator LF peak (grey), with the simulator LF peak mean marked (dots).

The right-hand panel of Figure 9 shows the additional analytical saturation-excursion contribution developed as a refinement to the ensemble-evaluator output on saturated cells. On the 13 non-saturated cells the contribution is at most a few centimetres and does not materially change the operator display; on bf8p5$_{q10w45}$ it adds 0.85 m to the reported $P_{95}$, keeping the red flag but not closing the underprediction. The refinement is therefore a marginal improvement in this regime, and the rejected-extensions section discusses why we did not extend it further.

## Rejected extensions

Three plausible refinements to the two-channel design were investigated and set aside.

**QTF-gradient estimation online.** Estimating the vessel's wave-drift QTF gradient at the operational sea state, then using it inside the ensemble evaluator to tighten the post-WCF envelope. Rejected on theoretical grounds: online QTF gradient estimation requires wave-field assumptions that break in operationally common conditions — mixed wind-sea and swell, cross-current, near-installation shielding — and the framework's measurement-only design goal (see Goal) does not accommodate a runtime wave-field model.

**Pre-trip variance as within-cell predictor.** Using the observed variance of the pre-WCF LF motion as a predictor of the seed-to-seed scatter in the post-WCF peak, within a fixed sea state. Empirically falsified: per-cell Pearson correlations between pre-trip variance and post-WCF peak range from -0.7 to +0.7 across the 14 cells of the roll-up, with no consistent sign.

**Pre-trip PSD-shape as within-cell predictor.** Same idea, but using the pre-WCF Vanmarcke spectral bandwidth. Empirically falsified on the same evidence, with the same range of correlations.

The two-channel design is therefore the **tight** envelope on this class of vessel and instrumentation: adding

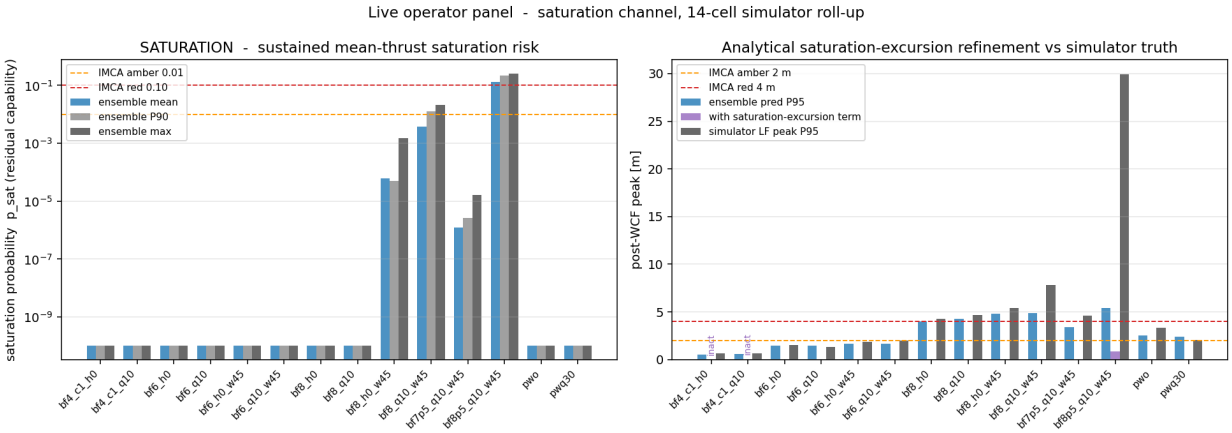



Figure 9: Saturation channel across the 14-cell simulator roll-up. Left: sustained mean-thrust saturation probability $p_{\text{sat}}$ per cell, ensemble mean / P90 / max shown against the amber (0.01) and red (0.10) thresholds. Right: the analytical saturation-excursion contribution on top of the ensemble-evaluator output, alongside simulator truth for reference.

observables from past history to characterise the future wave-phase-at-trip alignment does not add predictive power.

## Real-World Case

### Data source and scope

The framework is being validated against logged operations from an in-service DP vessel of opportunity. The vessel is a [TODO: vessel class, e.g. platform supply vessel] operating in [TODO: geographical area, e.g. North Sea], instrumented with the standard DP telemetry channels used by the live layer of the framework: position deviation, controller thrust command, allocator feedback, and DP model state (see the positioning section). The logs cover [TODO: N] operational periods totalling [TODO: N] hours across [TODO: sea-state / heading range summary].

The scope of the real-world validation is deliberately narrow: intact motion only. The framework's post-WCF prediction cannot be validated against real logs because no operator will intentionally trigger the worst-case failure for a paper. Post-WCF validation therefore stays with the simulator-truth roll-up of the simulation studies; the real-world case adds evidence that the intact-side machinery works when its input is the real world rather than a simulator.

### Method: forecast-based motion capability vs realised motion

The comparison is done in three steps.

1. For each logged operating window, the archived weather forecast in force at the start of the window is obtained from [TODO: forecast provider — ECMWF operational archive, MET Norway, NOAA, or a commercial subscription]. The forecast directional wave spectrum, mean wind, and mean current are fed through the pre-operation motion capability model of the method section to produce the predicted $P_{95}$ intact motion envelope for the vessel and, where applicable, for mission-equipment points.

2. The realised motion is computed from the vessel logs as the running maximum of $|\eta_{xy}t|$ over the 30-minute operating window, using the same definition the framework uses internally. Predicted vs realised $P_{95}$ is the primary comparison metric; the intact $P_{50}$ is reported as a secondary metric.
3. The archived forecast is cross-checked against the ERA5 [12] (or NORA3 [6] in the North Sea and Norwegian Sea) hindcast reanalysis. If the archived forecast was materially different from the hindcast for the same window, the comparison in step 2 is not a fair test of the framework and the window is flagged; the aggregate result is reported both including and excluding flagged windows.

### Status

Data acquisition and processing are in progress at the time of writing. The full set of predicted-vs-realised $P_{95}$ residuals across the [TODO: N] operational periods will be reported in the final version of the paper. If the comparison is not complete by the paper deadline, this section will present the methodology and initial results; the full outcome will be presented at the conference.

## Conclusion

This paper has developed a motion-based consequence quality analysis (CQA) framework that closes the motion gap in the classical DP capability and CQA machinery. The contribution is two-part. The pre-operation risk assessment (Motion-Based Capability Analysis) extends the classical capability analysis with an estimate of vessel and mission-equipment motion from the forecast sea state, delivered as the operability polar, the WCFDI decision matrix, and the pre-op operator summary. The live CQA panel (Consequence Analysis (Live)) reformulates the same underlying model for online execution, delivering intact and post-WCF motion $P_{95}$ continuously from real-time measurements and the DP model, with a saturation-probability trust indicator that fires when the linear motion prediction is no longer reliable.

The empirical basis is a 14-cell simulator validation across Beaufort 4 to Beaufort 8.5 (Simulation Studies): the intact-side $P_{95}$ is accurate to within +7% to +28% (systematically conservative by design), the post-WCF $P_{95}$ is within -23% to +16% on the eleven non-saturated cells, and the two-channel operator decision surface correctly red-flags the three saturated cells through two independent gates even where the motion prediction itself is understated. Real-vessel validation of the intact side against archived forecasts and hindcast is in progress (Real-World Case).

The framework preserves the operator's mental model — motion, footprint, proximity to limits — rather than translating the DP system's actions into force reserves that the operator then has to map back to expected motion. The classical Drift Off and Force Off alarms remain the primary hard-failure signals; the framework adds one motion-based alert (Large Transient) that tells the operator whether the motion forecast can be trusted as a tight upper bound or should be read as a lower bound.

The framework is deliberately additive to the classical rules. The capability + CQA computation mandated by IMO MSC/Circ.645, DNV-ST-0111 and IMCA M254 is not superseded or contested; applying a percentage capacity margin on top of the classical CQA continues to land the operator comfortably inside a safe operating envelope for the majority of operations. What the motion layer adds is the ability to work closer to the operational limit in a controlled way — the operator sees continuous, graded, motion-based proximity to the limit rather than waiting for a breach. This directly enables the walk-to-work and heave-compensated operational regimes that have expanded the DP envelope beyond simple hold-vs-drift.

The measurement-only online scope set out in the Goal section is honoured throughout: the live layer consumes only signals every DP-2/DP-3 vessel already publishes, plus vessel-constant parameters. No online

seakeeping model, no runtime spectral partition, no runtime wave-field assumption. This is a deliberate scope limit that keeps the design defensible under operationally common mixed sea states, cross-current forcing, and near-installation shielding effects that would break wave-field-based online refinements (see Rejected extensions).

Future work has three lines. First, extension of the mission- equipment coverage to further motion-sensitive systems (LARS, heave-compensated cranes, ROV winch handling). Second, completion of the real-vessel validation across a broader vessel population and operational area, building the field evidence needed to support rules-alignment discussions. Third, presentation of the motion-based CQA deliverables to the rulemaking bodies as candidates for inclusion in the next revision of IMCA M254 and the DNV-ST-0111 transient analysis guidance — a natural next step given that the current guidance already gestures in this direction (see Introduction) but stops short of requiring the delivery.

## Acknowledgements

**(TODO: whoever helped)**

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